# Mikhail Razakov

## Compiler / Software Protection Engineer

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Compiler/program-analysis engineer with professional LLVM/static-analysis experience and a historical Unity licensing project. Strongest inspectable signals are LLVM legality/transformations, x86-64 codegen/register allocation, and Roslyn data flow; protection experience is deliberately bounded to server-side key activation, activation limits, and Unity-client integration.

## EXPERIENCE

**July — August 2026**

### MCST — Compiler Engineering Intern

- Implemented a LICM pass for LLVM 22 in C++; hoisting conditions check loop structure, memory access, side effects, and speculative safety.
- Built a Python harness comparing program behavior before/after optimization and checking transform/no-transform cases.

**historical project · date not claimed**

### Unity licensing prototype

- Server-side key issuance/activation, activation-count limits, and Unity-client integration; no anti-debugging or anti-tamper claim.

**2026**

### Personal project — VpnMediator access-state backend

- Built server-side subscription/access state, device limits, payment flows, and recovery/rollback paths across failures and restarts.

## PROJECTS

### LLVM Interprocedural Global IV Optimization

An LLVM 22 pass localizes supported induction-like globals after interprocedural effects/affine analysis and a separate legality decision; APInt models fixed-width wraparound.

29 positive/negative regression cases: global-iv,verify, structural expectations, idempotence, and observable before/after execution comparison.

### x86-64 Compiler Backend & Register Allocation

A compiler backend with SSA/CFG validation, liveness/interference analysis, linear-scan and seeded simulated-annealing allocators, spills, phi lowering, and a SysV x86-64 emitter.

An independent assignment verifier plus native-vs-interpreter differential execution; tests cover branches, loops, spills, and parallel phi moves.

### DerefAfterNullAnalyzer

A Roslyn ControlFlowGraph analyzer propagates null state over conditional edges to a fixed point, handles loops/back edges, and merges states from multiple predecessors.

Handles method/property/index/event dereferences and invalidation after assignment/ref/out; dedicated analyzer tests use Microsoft.CodeAnalysis.CSharp.Testing.

## SKILLS

**Compiler / LLVM**
C++23, LLVM 22/IR, interprocedural analysis, legality, transformations, LLVM Verifier

**Program Analysis**
Roslyn CFG, fixed-point data flow, state propagation/joins, conservative analysis

**Low-Level / Codegen**
Rust, SysV x86-64, liveness/interference, register allocation, spills, phi lowering

**Software Protection / Licensing**
server-side activation, activation-count limits, Unity integration

## EDUCATION

HSE University — Software Engineering, 2026–2030

## RECOGNITION

**HSE Faculty of Computer Science — open-source projects competition**
Selected as one of the winners with UniversalToolchain / Wist2.

**LangDev'26 — Build the Language, Then Make the Abstractions Disappear**
Delivered the UniversalToolchain architecture talk at LangDev on 9 October 2026.

**Earlier distinctions**
MEPhI Junior: Information Technologies section winner in 2025 and first-degree diploma in 2026; HSE Vysshaya Proba Industrial Programming diploma holder; Baltic Science and Engineering Competition: first-degree diploma and the "Perfection as Hope" prize.

Moscow / remote